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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FAX (sources/stacks-chow/pinned-original-source.tex line 10436).

Allowed standard prerequisites (editorial).

- The localized classes P'_p and c'_p and the comparison with the restricted classes on E. Same chapter, Lemma labelled lemma-localized-chern-pre, cited at proof lines 10451-10462, and Lemma labelled lemma-silly-compose, cited at proof lines 10458-10462. Supplies the classes and the relation over E that the lemma transports to Z.
- The ring structure on localized classes. Same chapter, Remark labelled remark-ring-loc-classes and Remark labelled remark-res-push, cited at proof lines 10435-10438 and 10472-10478. Defines the products appearing in the polynomial identities.
- Pushforward of the restricted classes and the commutation with C. Same chapter, Lemma labelled lemma-silly, cited at proof lines 10484-10486, and Lemma labelled lemma-homomorphism, cited at proof lines 10488-10490. Rewrites the products through c_{p_i}(Q|_{W_infinity}) so that both sides coincide.
- The class C of the inclusion at infinity. Same chapter, Lemma labelled lemma-gysin-at-infty, invoked at proof lines 10449-10453. Provides the class through which P'_p and c'_p are defined.

Source reference. Original lemma 0F9K: sources/stacks-chow/pinned-original-source.tex, source lines 10204–10204. Same-chapter pointer into the author environment lemma at source line 10204; invoked at proof line 10452 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original reference 0FA0: sources/stacks-chow/pinned-original-source.tex, source lines 6419–6419. Same-chapter pointer into the author environment remark at source line 6419; invoked at proof line 10466 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line 10454 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9H: sources/stacks-chow/pinned-original-source.tex, source lines 9695–9695. Same-chapter pointer into the author environment lemma at source line 9695; invoked at proof line 10456 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FAK: sources/stacks-chow/pinned-original-source.tex, source lines 9867–9867. Same-chapter pointer into the author environment lemma at source line 9867; invoked at proof line 10458 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0FA1: sources/stacks-chow/pinned-original-source.tex, source lines 6451–6451. Same-chapter pointer into the author environment remark at source line 6451; invoked at proof line 10473 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FAV: sources/stacks-chow/pinned-original-source.tex, source lines 10353–10353. Same-chapter pointer into the author environment lemma at source line 10353; invoked at proof line 10483 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Lemma 0F9K assume $Q|_T$ is zero. In $A^*(Z \to X)$ we have*

$$\begin{aligned}
P'_1(Q) &= c'_1(Q), \\
P'_2(Q) &= c'_1(Q)^2 - 2c'_2(Q), \\
P'_3(Q) &= c'_1(Q)^3 - 3c'_1(Q)c'_2(Q) + 3c'_3(Q), \\
P'_4(Q) &= c'_1(Q)^4 - 4c'_1(Q)^2c'_2(Q) + 4c'_1(Q)c'_3(Q) + 2c'_2(Q)^2 - 4c'_4(Q),
\end{aligned}$$

and so on with multiplication as in Remark 0FA0.

Proof. The statement makes sense because the zero sheaf has rank < 1 and hence the classes $c'_p(Q)$ are defined for all $p \geq 1$. In the proof of Lemma 0F9K we have constructed the classes $P'_p(Q)$ and $c'_p(Q)$ using the bivariant class $C \in A^0(W_\infty \to X)$ of Lemma 0F9J and the bivariant classes $P'_p(Q|_E)$ and $c'_p(Q|_E)$ of Lemma 0F9H for the restriction $Q|_E$ of Q to the inverse image E of Z in W_∞. Observe that by Lemma 0FAK we have the desired relationship between $P'_p(Q|_E)$ and $c'_p(Q|_E)$. Recall that

$$P'_p(Q) = (E \to Z)_* \circ P'_p(Q|_E) \circ C \quad \text{and} \quad c'_p(Q) = (E \to Z)_* \circ c'_p(Q|_E) \circ C$$

To finish the proof it suffices to show the multiplications defined in Remark 0FA0 on the classes $a_p = c'_p(Q)$ and on the classes $b_p = c'_p(Q|_E)$ agree:

$$a_{p_1}a_{p_2}\ldots a_{p_r} = (E \to Z)_* \circ b_{p_1}b_{p_2}\ldots b_{p_r} \circ C$$

Some details omitted. If $r = 1$, then this is true. For $r > 1$ note that by Remark 0FA1 the multiplication in Remark 0FA0 proceeds by inserting $(Z \to X)_*$, resp.

$(E \to W_\infty)_*$ in between the factors of the product $a_{p_1}a_{p_2}\ldots a_{p_r}$, resp. $b_{p_1}b_{p_2}\ldots b_{p_r}$ and taking compositions as bivariant classes. Now by Lemma 0F9H we have

$$(E \to W_\infty)_* \circ b_{p_i} = c_{p_i}(Q|_{W_\infty})$$

and by Lemma 0FAV we have

$$C \circ (Z \to X)_* \circ a_{p_i} = c_{p_i}(Q|_{W_\infty}) \circ C$$

for $i = 2, \ldots, r$. A calculation shows that the left and right hand side of the desired equality both simplify to

$$(E \to Z)_* \circ c'_{p_1}(Q|_E) \circ c_{p_2}(Q|_{W_\infty}) \circ \ldots \circ c_{p_r}(Q|_{W_\infty}) \circ C$$

and the proof is complete. □